

TABLE I
SIGNAL LOGIC LEVELS AT DIFFERENT STAGES OF TRANSMITTER SIDE FOR A PARTICULAR COMMAND

Command sent from MATLAB GUI	Arduino output via digital pins (9, 10, 11, 12)	Encoder data input pins (AD8, AD9, AD10, AD11)
FORWARD	HLHL	HLHL
REVERSE	LHLH	LHLH
RIGHT	LHLL	LHLL
LEFT	LLLH	LLLH
STOP	LLLL	LLLL

TABLE II
SIGNALS LOGIC LEVELS AT DIFFERENT STAGES OF RECEIVER SIDE FOR A PARTICULAR COMMAND

Command sent from MATLAB GUI	Decoder data input pins (D8, D9, D10, D11)	L293D output pins OUT(1, 2, 3, 4)
FORWARD	HLHL	HLHL
REVERSE	LHLH	LHLH
RIGHT	LHLL	LHLL
LEFT	LLLH	LLLH
STOP	LLLL	LLLL

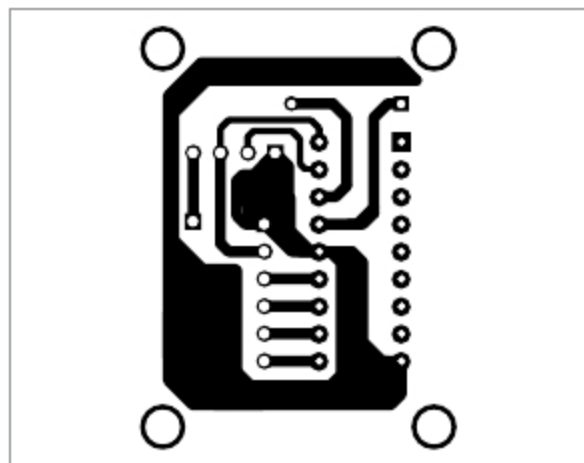


Fig. 8: Actual-size PCB layout of transmitter circuit

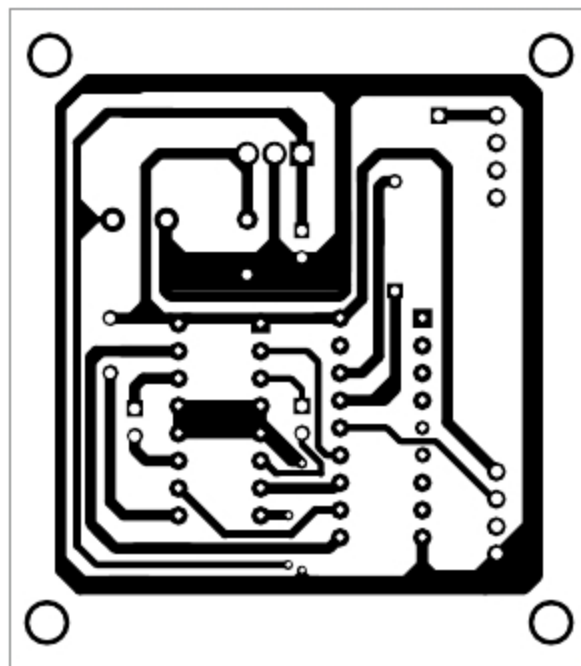


Fig. 10: Actual-size PCB layout of receiver circuit

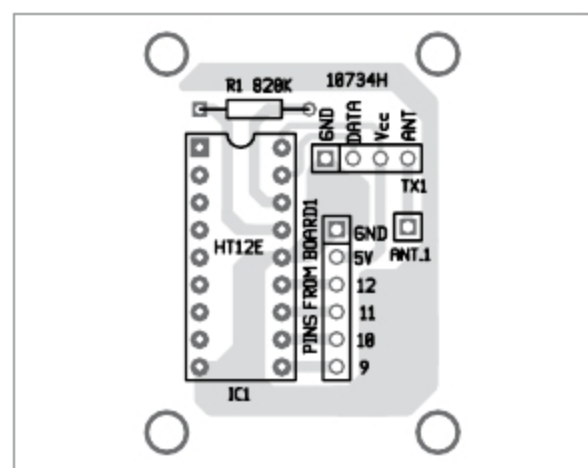


Fig. 9: Components layout for PCB layout in Fig. 8

controlling of the robotic car are listed in Tables I and II.

Construction and testing

An actual-size PCB layout for transmitter side is shown in Fig. 8 and its components layout in Fig. 9. An actual-size PCB layout of receiver side is shown in Fig. 10 and its components layout in Fig. 11.

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

Board1	- Arduino Uno board
IC1	- HT12E encoder
IC2	- 7805, 5V voltage regulator
IC3	- HT12D decoder
IC4	- L293D motor driver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 820-kilo-ohm
R2	- 47-kilo-ohm

Capacitors:

C1	- 100 μ F, 16V electrolytic
C2	- 0.1 μ F ceramic disk

Miscellaneous:

BATT.1	- 12V battery
M1-M2	- 12V DC geared motor, 50rpm
TX1	- 434MHz RF transmitter module
RX1	- 434MHz RF receiver module
ANT.1, ANT.2	- 17cm-long antenna wire
	- USB cable
	- 16-pin IC socket
	- 18-pin IC socket (2 No.)
	- 6-pin Berg male connector for Arduino
	- Heat-sink for 7805
	- Jumper wires
	- Wheels for motors
	- Chassis of robot

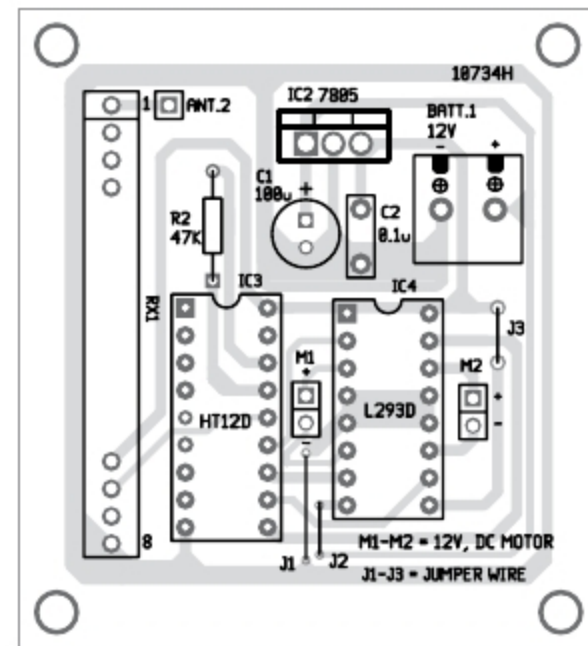


Fig. 11: Components layout for PCB layout in Fig. 10

After mounting the components on the transmitter PCB, connect the respective Arduino pins to the marked pins, as shown in Fig. 9.

Similarly, assemble the components on the receiver PCB. Mount the receiver PCB on the robot chassis. A 9V battery, as shown in the prototype, may not be powerful enough to drive the motors. It is recommended to use a 12V battery instead. Use suitable DC geared motors and wheels. **EFY**



Saikat Patra is passionate about electronics and MCU-based embedded system applications



Shibendu Mahata is M.Tech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. Currently, he is pursuing PhD from NIT Durgapur. He is interested in designing MCU-based real-time embedded signal processing and process control systems